```
count = 0
for name, lazy_tensor in as_dict.items():
    print(f"{name}: shape={lazy_tensor.shape} type={lazy_tensor.data_type}")
    if len(lazy_tensor.shape) == 1:
        count = lazy_tensor.shape[0]
    else:
        count = lazy_tensor.shape[0] * lazy_tensor.shape[1]

    if lazy_tensor.data_type == DT_F32:
        nBytes += count * 4
    else: #DT_F16
        nBytes += count * 2

    nParams += count

print(f'参数量={nParams}, 字节数={nBytes}')
```

代码运行输出：

```
model.embed_tokens.weight: shape=[32000, 4096] type=F16
model.layers.0.self_attn.q_proj.weight: shape=[4096, 4096] type=F16
model.layers.0.self_attn.k_proj.weight: shape=[4096, 4096] type=F16
model.layers.0.self_attn.v_proj.weight: shape=[4096, 4096] type=F16
model.layers.0.self_attn.o_proj.weight: shape=[4096, 4096] type=F16
model.layers.0.self_attn.rotary_emb.inv_freq: shape=[64] type=F32
model.layers.0.mlp.gate_proj.weight: shape=[11008, 4096] type=F16
model.layers.0.mlp.up_proj.weight: shape=[11008, 4096] type=F16
model.layers.0.mlp.down_proj.weight: shape=[4096, 11008] type=F16
model.layers.0.input_layernorm.weight: shape=[4096] type=F16
model.layers.0.post_attention_layernorm.weight: shape=[4096]
……
model.layers.23.self_attn.q_proj.weight: shape=[4096, 4096] type=F16
model.layers.23.self_attn.k_proj.weight: shape=[4096, 4096] type=F16
model.layers.23.self_attn.v_proj.weight: shape=[4096, 4096] type=F16
model.layers.23.self_attn.o_proj.weight: shape=[4096, 4096] type=F16
model.layers.23.self_attn.rotary_emb.inv_freq: shape=[64] type=F32
model.layers.23.mlp.gate_proj.weight: shape=[11008, 4096] type=F16
model.layers.23.mlp.up_proj.weight: shape=[11008, 4096] type=F16
model.layers.23.mlp.down_proj.weight: shape=[4096, 11008] type=F16
model.layers.23.input_layernorm.weight: shape=[4096] type=F16
model.layers.23.post_attention_layernorm.weight: shape=[4096] type=F16
参数量=4988274176, 字节数=9976551424
```

注意：这里只是一个 bin 文件的参数量和字节数，llama-2-7b 模型全部参数保存在两个 bin 文件中。

第6章

模型训练

